DATA SYSTEMS

PosPro

GNSS/INS trajectory post-processing

STAGE 01 OF THE DCS CHAIN · POSPRO → ALIGNPRO → CLOUDPRO

User Guide

For PosPro 1.8.0 · November 2026

Canadian Lidar · QGIS plugin, part of the Data Collection Systems chain. Needs the DCS Core plugin.

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1 Introduction

PosPro computes the trajectory of a Canadian Lidar mobile mapper: where the Certus GNSS/INS was, and how it was oriented, at every moment of a drive. It reads the mapper's captures (`.clraw`) or the Certus logs (`.anpp`) directly, combines the raw IMU and the raw satellite observations of both Certus antennas with base station data, and runs the inertial filter forward and backward over the drive. The two passes are combined into one smoothed trajectory, which you check on the map and in plots, clean up in Trajectory Review and export as SBET/SMRMSG, text, GeoPackage or GPX.

PosPro is **stage 01** of the DCS chain. Every time you process, PosPro also publishes the trajectory for the next stages: **AlignPro** uses it to solve the Lidar mounting (boresight), and **CloudPro** uses it to georeference the point cloud. You do not export anything by hand for them.

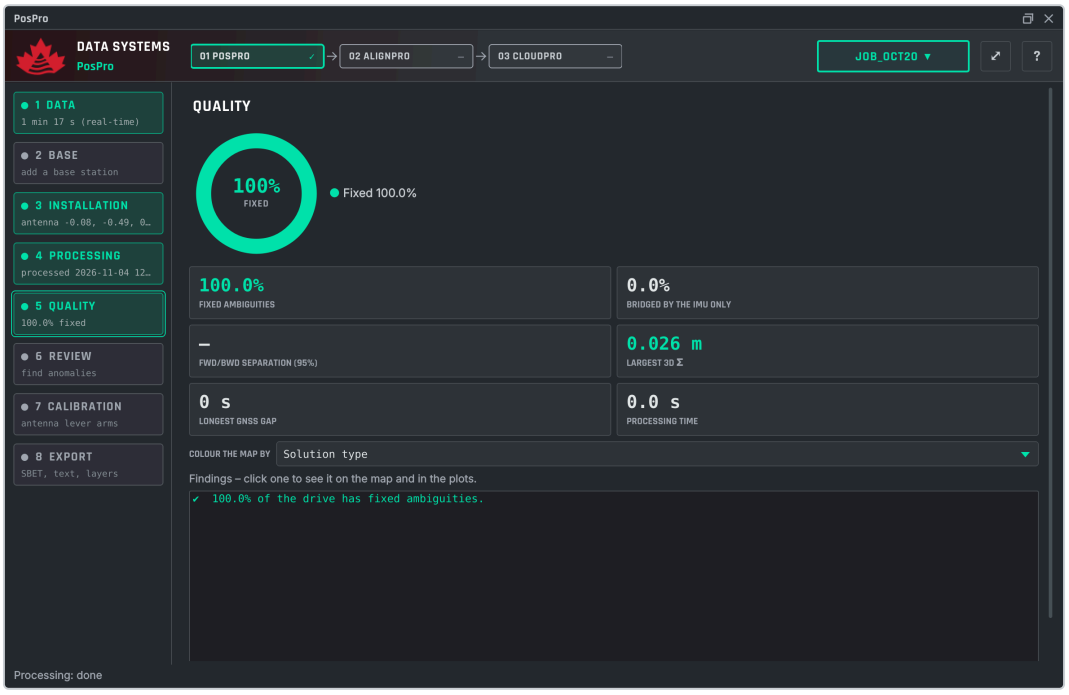


Figure 1.1 The PosPro window: the header with the workflow boxes and the project box, the eight steps on the left, the open step on the right, and the status line at the bottom.

What PosPro reads and makes

	DETAILS

Rover data	Mapper captures (`.clraw`, also unfinished `.clraw.partial` files) and Certus logs (`.anpp`). PosPro uses the Certus packets 20 (real-time solution and attitude), 28 (raw IMU), 60 (raw satellite observations), 61 (GPS ephemeris), 185 (installation) and 196 (dual-antenna set-up), and the mapper's `hardware.json` inside the capture.
Base stations	RINEX observation files (also Hatanaka-compressed and gzipped), or downloads from the base-station finder. Base positions from the RINEX header, typed in, built-in PPP or NRCan CSRS-PPP.
Processing	Tightly coupled GNSS/INS (both antennas), loosely coupled, GNSS only, or the Certus real-time solution. Differential (PPK) or precise point positioning (PPP). Forward and backward passes, combined.
Checks	Quality summary and findings, map layers coloured by solution type, accuracy or separation, quality plots, Trajectory Review, antenna lever-arm solver.
Output	SBET + SMRMSG, text files with the columns you choose, GeoPackage layer, GPX track, export note; the published trajectory for AlignPro and CloudPro.

Processing runs in a separate engine program (with RTKLIB inside), so a fault in processing never closes QGIS. The forward and backward passes run at the same time.

The workflow

Work down the eight steps on the left of the window. Each step shows its state in colour with a short note under its name:

- **Green**: done (for example *100.0% fixed, 3 reviewed*).
- **Amber**: needs a look (a base position from the RINEX header, an antenna offset of zero, review windows still open, fewer than 95 % fixed).
- **Red**: a problem (for example *no raw GNSS* when the captures hold no raw satellite observations and a method that needs them is chosen).
- **Grey**: not done yet; the note says what is missing.

1. **Data**: the rover files, the time range and the ephemeris.
2. **Base**: the base station and its position.
3. **Installation**: vehicle, motion sensor and the offsets of the antennas, the Lidar and the rear axle.
4. **Processing**: method and settings; `PROCESS`.
5. **Quality**: the summary, the findings, the map colours and the quality plots.
6. **Review**: find the parts that miss your limits and decide what to do with each.
7. **Calibration**: solve the antenna lever arms from the drive.
8. **Export**: SBET/SMRMSG, text, GeoPackage, GPX.

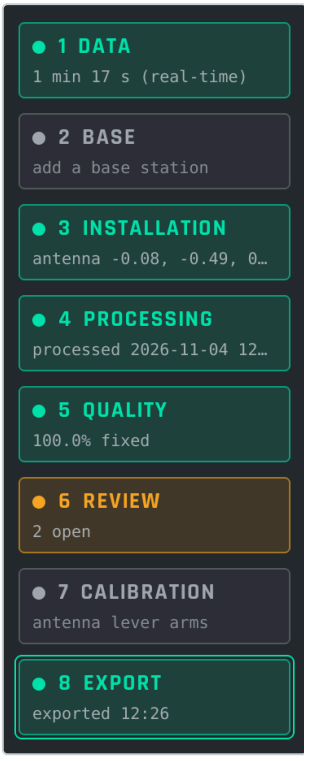


Figure 1.2 The steps and their states.

Steps 1 to 3 hold settings; they are saved as you change them. Step 4 runs the processing; steps 5 to 8 work on its result.

What's new

VERSION	CHANGES

1.8.0	**CHECK ANTENNAS** (Installation page and Trajectory editor): measures where GPS2 really was relative to GPS1 while driving straight, using the direction of travel rather than the Certus heading, and says whether the antennas match the installation or look swapped; one click then swaps the offsets and aligns from motion. Trajectory editor: **DIRECTION** chevrons along the track, **ANTENNAS** (GPS1 and GPS2 drawn along the track, and the vehicle at the point under the mouse) and the measured GPS2 positions. The QGIS map's trajectory and real-time lines show arrowheads in the direction of travel.
1.7.0	**Trajectory editor**: the trajectory in plan and in a height profile side by side, linked both ways – hover or click in one and the other shows the same moment, select bad points in either (rectangle, lasso, polygon, time range) and fix them there. **INTERPOLATE** now works in every processing mode, as a *smooth curve* (follows bends) or a *straight line*, horizontally and/or vertically, also at the start or end of the trajectory. New decision **REJECT** (bad, to fix later). See *Trajectory Review*.
1.6.3	When DCS Core is updated while QGIS is running, PosPro says to restart QGIS instead of asking for an update.
1.6.2	Installation: **IMU MOUNTING CORRECTION** – the pitch and yaw of the vehicle against the IMU, as a start for the estimate or held fixed, and **USE THE VALUES FROM THE LAST PROCESSING**. Trajectory Review: a **SHOW** filter and sorting by any column.
1.6.1	Navigation files next to the base observations, and files you add, are checked orbit by orbit against the Certus's own orbits and the IGS broadcast orbits before processing; orbits that disagree are left out and a warning names the file and the satellites (see *Data*). Processing stops with a clear error when the solution runs away (σ above 1 km) instead of handing over an unusable trajectory. Trajectory Review: **REMOVE FROM LIST** and **CLEAR HAND SELECTIONS**; one drawing on the map adds one window (or a few), not one per piece; hand selections from before are taken off the list when you process again, and the anomalies are found again on the new trajectory with your limits.
1.6.0	Guided steps: a **‹ BACK** / **NEXT ›** bar under the pages says whether each step is done and what to press, and **CONTINUE IN ALIGNPRO ›** on the last step once the trajectory is published (see *The PosPro window*). The version number is shown after the plugin name in the header. Fix (DCS Core 1.5.1): QGIS 4 on Windows no longer closes when you click in a floating window. Needs DCS Core 1.6.0.
1.5.3	Fixes from the first real field test: a cycle slip on a signal that returns after a short dropout is caught (it could pull a fixed solution metres off); short losses of IMU data are bridged and the Data step reports how much is missing; the offset between the vehicle and the IMU's aligned frame is estimated while driving and reported; tightly coupled passes start from the fixed GNSS-only position. On the test drive: under 35 % fixed before, 99.8 % now. New: Recording settings.
1.5.2	Fix: no more "unclosed file" warnings in the QGIS log after processing.
1.5.1	Mapper 2.10 projects: a new project offers the navigation recordings (recommended) or the survey lines; rover files are read in recording order; a navigation recording's hardware record is completed from a survey line. Needs DCS Core 1.4.0.
1.5.0	The same header as AlignPro and CloudPro: logo, the plugin name, the workflow boxes beside it, the project box on the right, then the Window button and the user guide button. The window title is just *PosPro*. The vehicles are the mapper's (Van, Truck, Car) and set the wheel constraints; the separate motion-constraints list is gone (older projects keep what they processed with). Captures without raw IMU take their time span from the real-time solution, and a real-time trajectory is published with the frame of the Certus RTK corrections and its own rate. This guide, rewritten with screenshots of this version. Needs DCS Core 1.3.0.
1.4.0	Named just PosPro. Opens in DCS Core's standard window: floating the first time, docked only when you choose it from the Window menu; dragging never docks it. It comes back as you left it, also after QGIS restarts; **RESET WINDOW** returns to the default. The quality plots open in the same kind of window. Neutral graphite look with lighter view backgrounds.
1.3.x	Trajectory Review (step 6). The real-time mode (the Certus solution, no base station). GPX export. The user guide is installed with the plugin.
1.2.x	Part of the DCS chain (needs DCS Core): projects in the job folder, the trajectory published for AlignPro and CloudPro, workflow boxes with out-of-date results, relocating a moved job folder, saving the QGIS project in the project folder. Older `.pospro.json` projects still open.

1.1.x	Tight coupling with both antennas, atmosphere estimation, signal-strength weighting, lever-arm solver, base-station finder, Installation filled from the capture, data coverage bars, `MEASURE GPS2 FROM THE DATA`.

2 Installing and opening PosPro

PosPro is one of three QGIS plugins that process a Canadian Lidar mobile mapper job one after the other. All three need the **DCS Core** plugin, which holds what they share: the job's processing project, the files each stage hands to the next, the look, the header and the window.

STAGE	PLUGIN	READS	PUBLISHES
01	**PosPro**	The mapper captures (`.clraw` or `.anpp`), base station data	The trajectory: `trajectory.json` with SBET and SMRMSG
02	**AlignPro**	The captures and PosPro's trajectory	The Lidar mounting: `alignment.json`
03	**CloudPro**	The captures, the trajectory and the mounting	The point cloud: `cloud.json`, LAS 1.4, ground grid, report

Requirements

- QGIS 3.34 or later, including QGIS 4.
- DCS Core 1.6.1 or later. PosPro 1.8.0 will not start with an older DCS Core: a message bar asks you to update it.

Install from the Canadian Lidar repository

1. In QGIS open `PLUGINS › MANAGE AND INSTALL PLUGINS › SETTINGS`, press **Add...** and enter the repository URL `https://canadian-lidar.github.io/qgis-plugins/plugins.xml`.
2. Press **Reload all Repositories**, then install **DCS Core** first and **PosPro** after it.
3. Restart QGIS. Later versions appear under `UPGRADEABLE` in the same window; update DCS Core together with the plugins.

Without internet access, use `INSTALL FROM ZIP` in the same window with `dcs_core-1.6.1.zip` first, then the PosPro zip.

Opening PosPro

- `PLUGINS › CANADIAN LIDAR › DATA SYSTEMS › POSPRO`, or the PosPro button on the Plugins toolbar.
- From another DCS plugin: click the **PosPro** box in its header to open PosPro on the same project.
- This guide: the `?` button in the header, `USER GUIDE (THIS VERSION)` in the project box, or `HELP › PLUGINS › POSPRO USER GUIDE`. The guide that opens always matches the installed version.

3 The PosPro window

PosPro, AlignPro and CloudPro share the same header and the same kind of window, so what you learn here applies to all three.

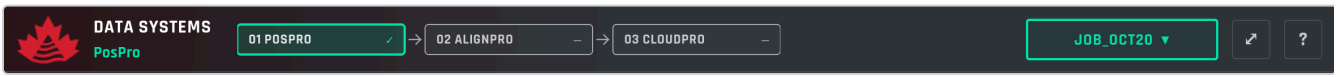


Figure 3.1 The header. Left: the logo and the plugin name, then the workflow boxes. Right: the project box, the Window button and the user guide button.

The header

PART	WHAT IT DOES
Plugin name and version `PosPro v1.6.0`	The plugin and the version you are running. Quote it when you report a problem; the user guide (`?`) is the one for this version.
Workflow boxes `01 POSPRO → 02 ALIGNPRO → 03 CLOUDPRO`	The state of each stage on the open project: ✓ published and current, ⚠ **stale** (out of date: an earlier stage was re-run, the raw data or a shared setting changed), **running**, **failed**, or – not run yet. Hover a box for the reasons. Click another stage's box to open that plugin on the same project. The boxes refresh every few seconds.
Project box	The open project's name; click it for the project menu (Projects). It turns red while PosPro is working.
⤢ Window	The Window menu: Floating window, Maximise, Dock right / left / bottom / top, Reset window, Close.
`?`	Opens this guide.

When the window is narrower than about 1,080 pixels (usually when it is docked) the workflow boxes move to a second row under the header.

Moving through the steps

Under the step pages a bar always says where you are and what comes next:

- `‹ BACK` goes to the previous step.
- The sentence in the middle says whether the step is **done** (✓, green), needs a **check** (⚠, amber: you can continue), is **not done yet** (○, with the button to press), or is **optional**.
- `NEXT: <STEP> ›` goes to the next step. It is greyed out until a required step is done; on optional steps it reads `SKIP TO <STEP> ›`.
- On the last step it becomes `CONTINUE IN ALIGNPRO ›` (from PosPro) or `CONTINUE IN CLOUDPRO ›` (from AlignPro). It opens the next plugin on the same project once this one has published its results; until then it says what is missing.

The step list on the left still opens any step at any time; the bar only guides. CloudPro, an editor rather than a list of pages, shows its three steps (Georeference, Clean & classify, Publish) in a strip under the header with one `NEXT` button that does the next thing.

Floating or docked

PosPro opens as a **floating window** the first time, centred on the QGIS screen, with its own minimise, maximise and close buttons. It docks into the QGIS window only when you choose to:

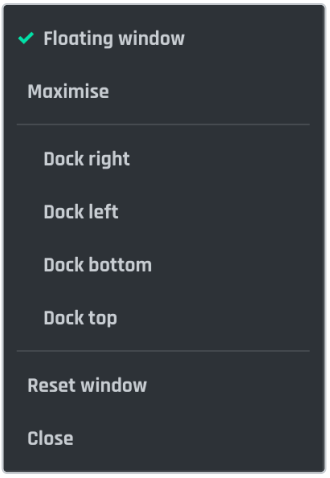


Figure 3.2 The Window menu, from the ⤢ button or a right-click on the title bar.

- **Dock right, left, bottom or top** puts it on that side of the QGIS window. Dragging the window never docks it, so it cannot snap into a corner by accident.
- **Floating window** takes a docked window out again; double-clicking the title bar of a docked window does the same.
- **Maximise** fills the screen; choose it again (it reads *Restore window size*) to go back.
- **Reset window** returns to the default floating window, centred, at its default size.

PosPro remembers how you left it (floating with its place and size, maximised, or docked with its width or height) and opens the same way next time, also after QGIS restarts.

Below the steps, the bottom of the window holds the **status line** (the last message, or the stage of the job that is running), a progress bar and a red `STOP` button while a job runs, and short **messages** (green, amber or red edge) that disappear after a few seconds. Warnings and errors also go to the QGIS message bar and to `SHOW THE LOG`. Only one job runs at a time; starting another while one runs gives *Wait for the current job to finish (or cancel it).*

The default window is 1400 × 950 pixels; docked left or right it is 720 pixels wide.

4 Projects and the job folder

PosPro, AlignPro and CloudPro work on one shared **processing project** per job. The mapper's capture folders are never changed: everything the plugins make goes in a `Processing` folder.

PATH	HOLDS
`Job_Oct20/`	The job folder: the capture folders recorded by the mapper (read only)
`Job_Oct20/Processing/<Name>/<Name>.dcsproj`	The shared project: captures, shared settings (coordinate system, base station, antennas, Lidar mounting, vehicle) and the state of each stage. A `.bak` copy is kept beside it.
`01_PosPro/`, `02_AlignPro/`, `03_CloudPro/`	Each stage's own settings file, `work/`, `log/`, and `out/` with what it publishes
`<Name>.qgz`	The QGIS project, when you save it from the project menu

Mapper 2.10 projects

From mapper 2.10 the mapper keeps its own projects. Use the mapper project's folder as the job folder:

PATH	HOLDS
`Deck_Test/project.json`	The mapper project: name, id, next line, operator, vehicle
`Deck_Test/lines/Deck_Test_L001_20261104_124512/`	One survey line (one Start/Stop), its files named `Deck_Test_L001_20261104_124512_0001.clraw`, `_0002` … Captures made before 2.10 that the mapper grouped into the project keep their old names here.
`Deck_Test/nav/Deck_Test_NAV_20261104_120000_0001.clraw`	A navigation recording: everything the Certus sent while the project was open (no Lidar). A new one starts at each power-on, project opening, UTC date change and every 2 GB.

If you pick the project's `lines` or `nav` folder, or a line folder, the New project dialog takes the mapper project's folder as the job folder and its name as the project name. The survey lines are recognised from what the files record (their line number and capture), not from their names, so a line split over several files is one line and old and new captures mix freely, in recording order. Navigation recordings are never taken as survey lines.

The Processing folder can also live on another drive (choose **Elsewhere…** when you create the project). Paths are stored relative to the job folder where possible, so a job can be copied or moved as a whole.

The project menu

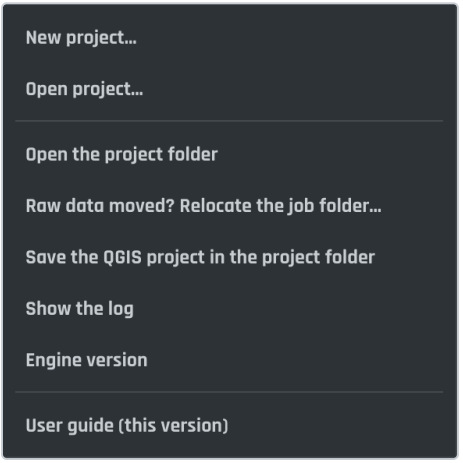


Figure 4.1 The project box menu in PosPro.

ITEM	WHAT IT DOES
NEW PROJECT…	Opens the New DCS project dialog (below). PosPro then offers the captures it finds in the job folder as rover files.
OPEN PROJECT…	Opens a processing project (`.dcsproj`), or an older stand-alone PosPro project (`.pospro.json`). The last project you had open reopens by itself when PosPro starts.
OPEN THE PROJECT FOLDER	Shows the project folder (`Processing/<Name>`) in the file manager.
RAW DATA MOVED? RELOCATE THE JOB FOLDER…	Asks for the job folder's new location and finds the captures there again; PosPro's rover file list follows. Not available for older stand-alone projects.
SAVE THE QGIS PROJECT IN THE PROJECT FOLDER	Saves the open QGIS project as `<Name>.qgz` in the project folder (asks first when the QGIS project already has another file name).
SHOW THE LOG	The PosPro log of this session: every message, warning and error with its time.
ENGINE VERSION	The version of the processing engine and of RTKLIB inside it, and where the engine is installed.
USER GUIDE (THIS VERSION)	Opens this guide.

Creating a project

1. Choose NEW PROJECT… in the project box.
2. **Job folder**: the folder that holds the mapper's capture folders. The project name defaults to the folder's name.
3. **Processing folder**: inside the job folder by default; **Elsewhere…** to put it on another drive.
4. Press OK. The project is created and opened. PosPro then looks through the job folder (five folder levels deep, skipping the Processing folder and the mapper's processed folders) for mapper captures: the survey lines' `.clraw` files and the mapper 2.10 navigation recordings, or the Certus logs (`.anpp`) when there is no `.clraw`, and asks whether to use them as the rover files (Figure 4.3). When a mapper 2.10 project holds both navigation recordings and survey lines, it asks which (Figure 4.4).

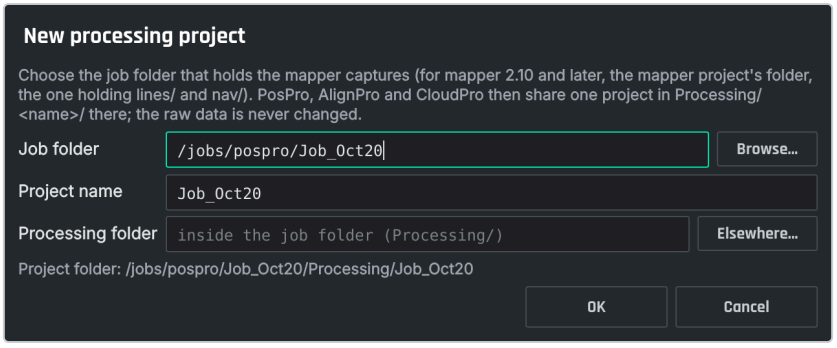


Figure 4.2 The New DCS project dialog (the same in every plugin).

A project made in any of the three plugins is the same project: open the `.dcsproj` in another plugin, or click its box in the header.

When the raw data has moved

If the captures are not where they were (another drive letter, a copied job), PosPro says so when the project opens and asks for the job folder's new location; `RAW DATA MOVED? RELOCATE THE JOB FOLDER…` in the project menu does the same at any time. The captures are found again by their place in the job folder; nothing goes out of date because of the move.

Out-of-date results

Each stage records what it was built from. When an earlier stage publishes again, the raw data changes, or a shared setting it depends on is edited, the stage turns ⚠ **stale** in the workflow boxes, with the reason on hover. Run that stage again to bring it up to date. A run that fails or is cancelled keeps the earlier results.

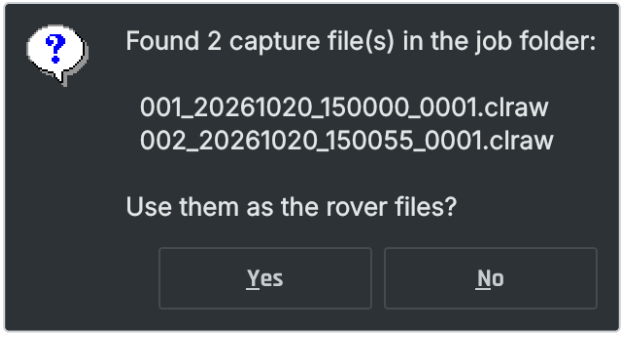


Figure 4.3 After a new project is created, PosPro offers the captures it found in the job folder as rover files.

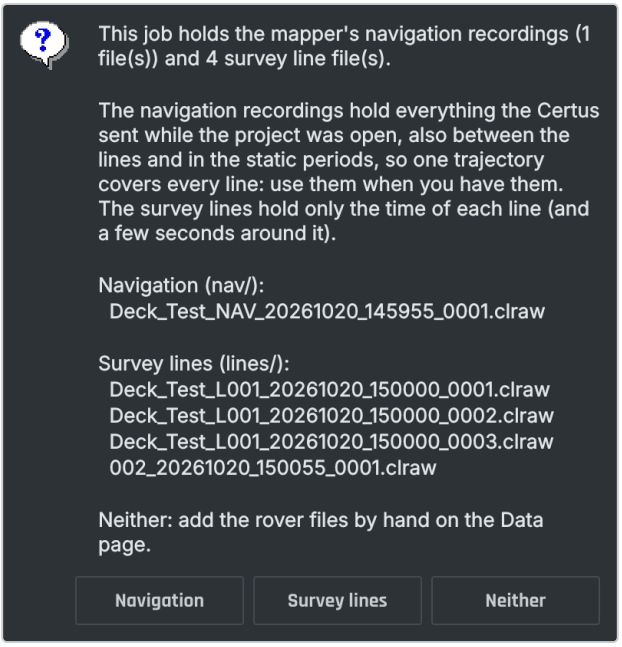


Figure 4.4 A mapper 2.10 project with a navigation recording and survey lines: choose **NAVIGATION** (recommended), **SURVEY LINES** or **NEITHER**.

Navigation or survey lines? A mapper 2.10 navigation recording holds everything the Certus sent while the mapper project was open: the static periods before and after the drive and the time between the lines as well as the lines themselves. Processed once, it gives one trajectory that covers every line, which suits PPK and tight coupling best. The survey lines hold only each line and a few seconds around it. Files are read in recording order, whatever their names. The navigation recording's hardware record has no vehicle, rear axle or coordinate system; PosPro takes those from a survey line of the same job.

PosPro keeps its own settings in `01_PosPro/pospro.json` and its work in `01_PosPro/work/` (extracted data, intermediate files, results). Shared facts that the other stages need (coordinate frame, base station, antenna offsets, vehicle, the capture's hardware record, the list of captures) are written into the shared project whenever you change them in PosPro.

5 Step 1 - Data

The Data page holds the rover files, what they contain, the time range to process and the satellite ephemeris.

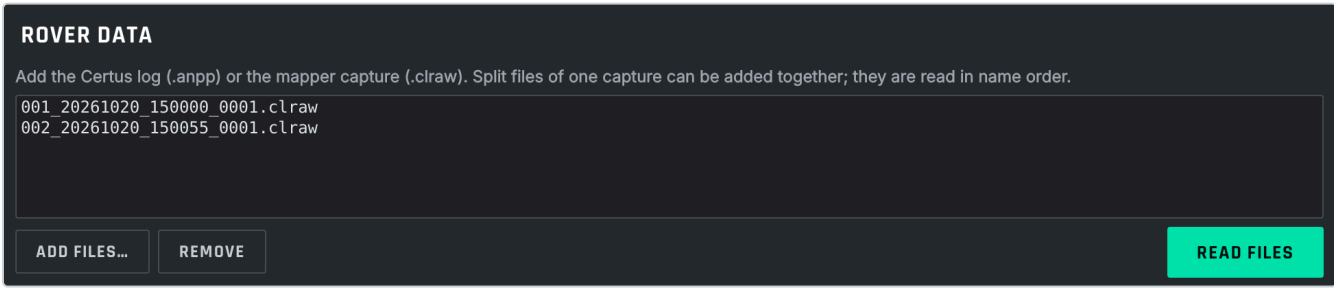


Figure 5.1 The rover files of the project: the two captures found in the job folder.

Rover data

- **ADD FILES...** adds Certus logs (`.anpp`) or mapper captures (`.claw`, `.claw.partial`). Split files of one capture can be added together; they are read in name order. Adding files clears the time range and reads the files at once.
- **REMOVE** takes the selected files off the list (the files themselves are not touched).
- **READ FILES** reads all files again. Reading extracts the raw IMU, the raw observations of both antennas, the real-time solution, the ephemeris and the capture's hardware record into the project's work folder. PosPro reads again by itself before processing whenever a file has changed.

A file that is no longer where it was shows in red (*File not found* on hover). After reading, PosPro fills the Installation offsets from the capture if they are still empty, with the message *Installation offsets filled from the capture (...)*.

What the files contain

After reading, this box lists what PosPro found. Lines in amber point at something to check.

LINE	SHOWS
Recording	Start and end (UTC) and duration of the IMU data; for captures without raw IMU, of the Certus real-time solution (the line says so).
IMU (packet 28)	Number of samples, rate and gaps (amber when there are gaps). Notes when samples before GNSS time was valid were skipped, or when IMU time stamps were shifted to match packet 20.
GNSS (packet 60)	Epochs, interval and the mean number of satellites, then the signals recorded. Amber notes: epochs not on whole seconds (base data are then interpolated; set the Certus to 125 ms for whole-second epochs), or GLONASS off on the Certus. *none – raw observations were not recorded* when the captures hold no raw observations; the Data step then turns red (*no raw GNSS*) and only the real-time mode can run.
Ephemeris (packet 61)	The number of GPS ephemerides the Certus recorded.
Checksum errors	Shown only when packets failed their checksum.
Mapper hardware	*hardware.json found in the capture*: the mapper's offsets can be used on the Installation page.

Time range to process (UTC)

`START` and `END` limit the processing to part of the capture, for example to leave out a long wait at the start. They cannot go beyond the recording. `WHOLE CAPTURE` clears the limits. The same range is used by `DATA-PAGE TIME RANGE ONLY` on the Export page.

Data coverage

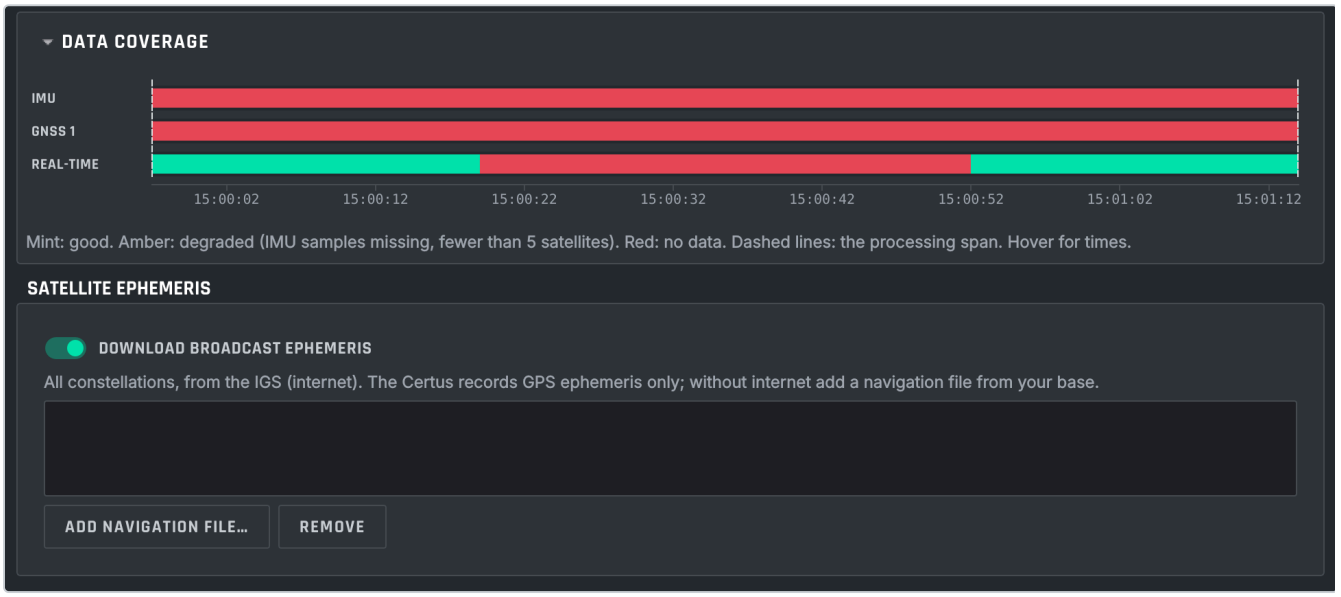


Figure 5.2 Data coverage and the ephemeris settings. Here the captures hold the Certus real-time solution only: the IMU and GNSS 1 bars are red and the real-time bar shows the two passes.

One bar per data stream over the time of the capture: **IMU**, **GNSS 1**, **GNSS 2** (when the second antenna was recorded) and **Real-time**. Mint is good, amber degraded (IMU samples missing, fewer than 5 satellites), red no data. The real-time bar takes the colour of the Certus fix type (RTK fixed mint, RTK float blue, DGNSS or SBAS amber, 2D/3D red). Dashed lines mark the processing span; hover a bar for times. The Base page shows the same bars with one more per base station.

Satellite ephemeris

- `DOWNLOAD BROADCAST EPHEMERIS` (on by default): merged broadcast ephemeris of all constellations from the IGS archives, for each day of the drive. Downloads are cached in the `pospro_cache` folder of your QGIS profile.
- The Certus records GPS ephemeris only. Without internet, add a RINEX navigation file from your base with `ADD NAVIGATION FILE…`; `REMOVE` takes it off the list. Navigation files lying next to the base observation files (same 4-character station name) are also used automatically.
- Every navigation file that is not the Certus's own or the IGS broadcast file is checked first: each orbit is compared with the Certus's orbits and, with internet, the IGS broadcast orbits of the same satellite. Orbits more than 30 m apart are left out, and the log names the file and the satellites. One bad orbit is enough to ruin a trajectory, so if the warning comes up, take the file out of the base folder. Satellites with no reference orbit (GLONASS without internet) cannot be checked and are kept.

6 Step 2 - Base stations

Differential processing (PPK) and tight coupling need one base station that covers the whole drive. Every point of the trajectory inherits the base position's error and its frame, so give the base a precise position before processing. The Base page is not needed for PPP or for the real-time mode (the step then reads *not needed (PPP)* or stays grey).

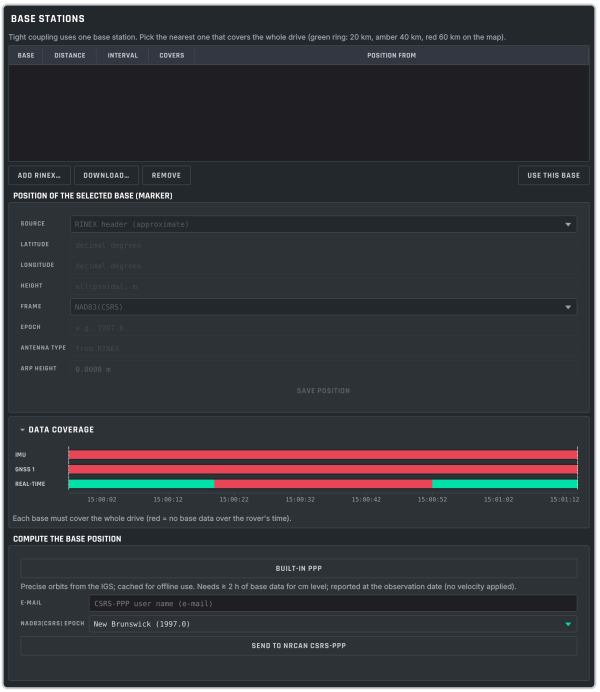


Figure 6.1 The Base page before a base station is added. The position fields are filled in once a base is selected in the table.

The base table

COLUMN	SHOWS
Base	The station name. ★ marks the base used for processing.
Distance	From the drive: green up to 20 km, amber up to 40 km, red beyond.
Interval	The observation interval of the file.
Covers	The share of the rover's time that the base file covers. It should be 100 %.
Position from	RINEX header, entered, built-in PPP or CSRS-PPP, with the frame and epoch.

- **ADD RINEX…** adds a base from RINEX observation files (`.obs .rnx .crx .??o .??d .gz`). Select several files of the same station to add them as one base; their observations are merged (they must list the same observation types, otherwise add them as separate bases). Compact (Hatanaka) RINEX is decompressed automatically.
- **DOWNLOAD…** opens the base-station finder (below).
- **REMOVE** removes the selected base; **USE THIS BASE** makes it the one used for processing (the first base added is used until you choose another).

The bases also appear on the QGIS map, in the group *PosPro: <project>*: the base used as a large red triangle, the others grey, with distance rings at 20, 40 and 60 km.

Position of the selected base (marker)

The coordinates are those of the marker. Choose the **SOURCE** (RINEX header (approximate), Entered / published, Built-in PPP, NRCan CSRS-PPP), enter **LATITUDE** and **LONGITUDE** in decimal degrees and the ellipsoidal **HEIGHT**, the **FRAME** (NAD83(CSRS), ITRF2020 / IGS20, WGS 84, NAD83(2011)) and its **EPOCH**, the **ANTENNA TYPE** (from the RINEX header; used with the IGS antenna calibrations) and the **ARP HEIGHT**, then press **SAVE POSITION**. The frame and epoch of the base become those of the project: the trajectory and every export are in that frame.

Compute the base position

BUTTON	USE WHEN	RESULT
BUILT-IN PPP	No CSRS-PPP account, or offline once the products are cached	Static PPP with downloaded precise orbits and clocks (ESA, IGS or CODE; finals first, then rapids) and the IGS antenna file. ITRF2020 at the observation date, transformed to NAD83(CSRS) at the same date when the base frame is NAD83(CSRS); no velocity applied. Needs at least 2 hours of base data for centimetre level; PosPro warns when the file is shorter.
SEND TO NRCAN CSRS-PPP	Internet and an NRCan account	Enter the **E-MAIL** you use as your NRCan user name and choose the **NAD83(CSRS) EPOCH**: New Brunswick (1997.0), Federal (2010.0) or Observation date. PosPro sends the file, waits for the result and enters the NAD83(CSRS) position at that epoch.

The result of each computation is listed under the buttons (built-in PPP: date, hours of data, products and σ). For work tied to NAD83(CSRS) epoch 1997.0, use CSRS-PPP with that epoch: the built-in PPP gives the position at the observation date, which differs by the crustal motion since 1997.

> APPROXIMATE BASE POSITIONS The Base step turns amber while the base position comes from the RINEX header, and **PROCESS** asks before using it: *The base position comes from the RINEX header and may be metres off; every point of the trajectory inherits that error.*

The base-station finder

DOWNLOAD… opens *PosPro – base stations near the drive*: a map of the reference stations within 150 km of the drive, with the drive itself in red and dashed rings at 20, 40 and 60 km. Read the rover files first; the map centres on the drive. The finder needs the internet.

NETWORK	SYMBOL	DOWNLOAD FROM POSPRO
NRCan CACS, NS active control, NRCan regional	squares (red, pink, dark red)	Yes, free
IGS	mint star	Yes, free
NOAA CORS	blue triangle	Yes, free
Can-Net, SmartNet, BrandtNet, TopNET, GEODNET, Other RTK	diamond, pentagon, hexagon, triangle, circle, cross	Through the network's portal (**OPEN PORTAL**, then **IMPORT FILE…**), or with your account where the network allows it

1. Untick networks in the legend under the map to hide them. Large symbols are the three nearest stations of each network; the table lists them with **NETWORK**, **STATION**, **KM**, **RATES** and **ACCESS** (Free download, Your account, Portal only).
2. Click a station on the map or a row. The card shows its name, operator, distance (green within 20 km, amber within 40, red beyond), position, receiver and antenna, status (for the commercial networks, NRCan's check of

their tie to NAD83(CSRS)), rates and access, and the days of the drive.

3. Choose the **RATE** (rates under 5 s come as 15-minute files, others as daily files), leave **PRECISE ORBITS, CLOCKS AND ANTENNA FILE** on, and press **DOWNLOAD**. A progress card shows the file, the megabytes and the time, with **CANCEL**. Networks that need your account ask for it once and keep it in the QGIS password store.
4. When the download finishes the finder closes and the station is added to the base table, with a message saying where the files were saved (the project's `work/base` folder).

Station lists are cached for a week; **REFRESH STATION LISTS** loads them again. The commercial stations come from NRCan's list of networks tied to NAD83(CSRS), so they cover Canada only.

7 Step 3 - Installation

Offsets are in metres from the Certus reference point, in the vehicle frame: **Forward +, Starboard +, Down +**. This is the same frame and the same values as the mapper's *Settings › Hardware*, and the page is laid out the same way. The offsets are filled from the capture when the rover files are read.

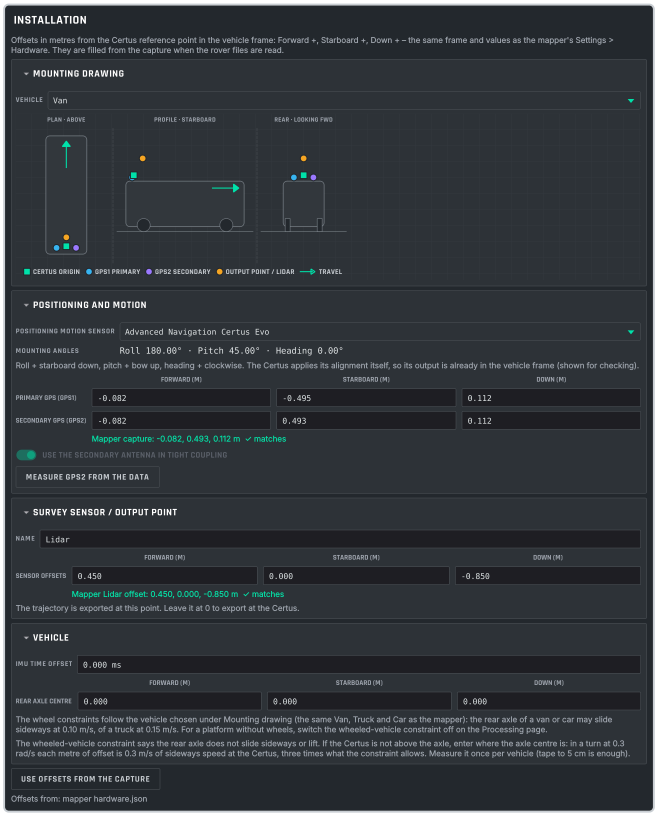


Figure 7.1 The Installation page, filled from the mapper's `hardware.json` in the capture. The green notes confirm that the values match the capture.

Mounting drawing

Choose the **VEHICLE** : Van, Truck or Car, the same vehicles and sizes as in the mapper (*Settings › Hardware*). When the rover files are read, the vehicle recorded in the capture is chosen for you. The vehicle also sets the wheel constraints used in processing (below). The drawing shows the vehicle from above, from starboard and from the rear, with the Certus origin, GPS1, GPS2, the output point (Lidar) and the rear axle as you enter them, so a sign error or a swapped axis shows at a glance.

Positioning and motion

- **POSITIONING MOTION SENSOR** : the IMU model, which sets the noise and bias values the filter uses: Advanced Navigation Certus Evo (default), Advanced Navigation Certus, Generic tactical-grade MEMS, Generic industrial MEMS.
- **MOUNTING ANGLES** : the Certus alignment (roll + starboard down, pitch + bow up, heading + clockwise) from the capture. The Certus applies its alignment itself, so its output is already in the vehicle frame; the angles are shown for checking only.
- **PRIMARY GPS (GPS1)** and **SECONDARY GPS (GPS2)** : the antenna offsets. Under each row PosPro shows what the Certus or the capture reports: green ✓ *matches* within 1 cm, amber ≠ *differs by … mm* otherwise. With the Certus dual antenna in automatic mode there is no GPS2 offset to read, and PosPro says so.

- `USE THE SECONDARY ANTENNA IN TIGHT COUPLING` (on by default): with both antennas, heading stays observable when parked or driving straight, and GNSS gaps are bridged better. It is greyed out when the captures hold no observations of the second antenna. GPS2 is used only when its offset is not zero.
- `MEASURE GPS2 FROM THE DATA`: computes the GPS1–GPS2 carrier-phase baseline from the recorded observations, turns it into the vehicle frame with the Certus attitude and adds it to GPS1. It works parked or driving, needs a few minutes with both antennas in the open, and asks before using the result (with its ± values, the antenna separation and the share of fixed epochs). Enter GPS1 first. If the antennas may be plugged in the wrong way round, run `CHECK ANTENNAS` first: swapped antennas also turn the Certus heading 180°, which would hide the swap from this measurement.
- `CHECK ANTENNAS`: see below.

Checking the antennas

If the PRIM and SEC antenna cables were plugged in the wrong way round, GPS1 (the primary receiver) is really on the secondary antenna, and every offset is wrong by the whole distance between the antennas. The Certus does not notice: its own heading simply turns 180°. `CHECK ANTENNAS` finds it from the recorded data, without a base station:

- It measures the GPS1 → GPS2 baseline from the carrier phase of both receivers (fixed epochs only), and while the vehicle drives straight above 11 km/h it puts that baseline in the vehicle frame using the **direction of travel**, which does not depend on the antennas.
- It compares the result with the offsets above: *Antennas as installed* (green), *Antennas swapped* (red: GPS2 is where GPS1 should be) or *Not conclusive* (amber: neither, or too little straight driving; parked data cannot tell, because then only the Certus heading is available).
- It also reports how far the Certus heading was from the direction of travel; about 180° confirms swapped antennas.

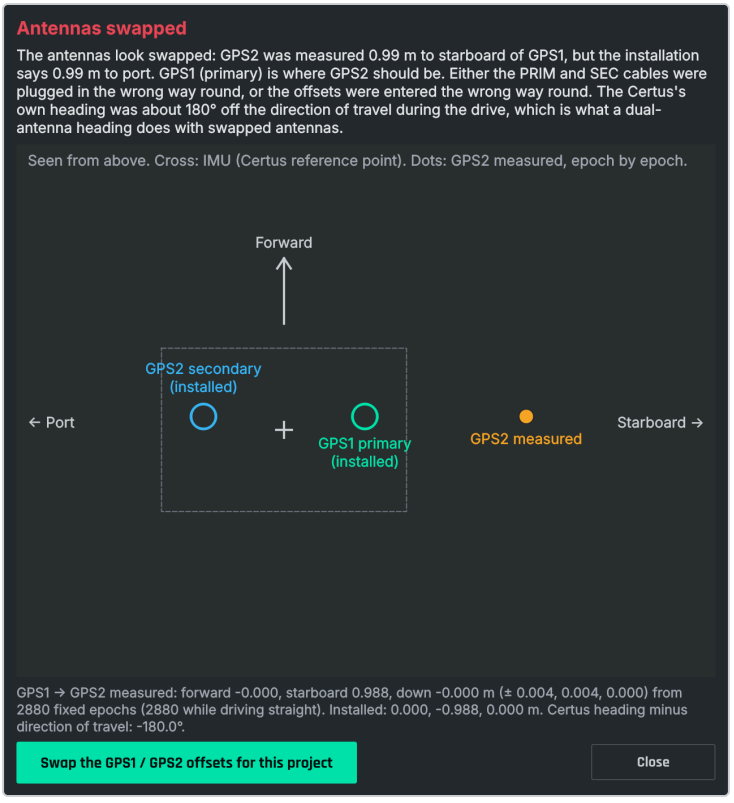


Figure 7.2 A drive recorded with the antenna cables swapped: GPS2 was measured 0.99 m to starboard of GPS1 instead of 0.99 m to port, and the Certus heading was 180° off.

When the antennas look swapped, **SWAP THE GPS1 / GPS2 OFFSETS FOR THIS PROJECT** processes the data as it was recorded: GPS1 gets GPS2's offset and the other way round, and when the Certus heading was turned 180° the INS is aligned from the direction of travel instead (Processing › Alignment *From motion*). Process again afterwards, and fix the cables on the unit for the next captures. In the Trajectory editor the measured GPS2 positions are drawn along the track as well (see *The Trajectory editor*).

Survey sensor / output point

A **NAME** (for example *Lidar*) and the **SENSOR OFFSETS** of the point you may want the trajectory exported at. It appears as a choice under **POSITION OF** on the Export page. Leave it at 0 to export at the Certus. The published trajectory for AlignPro and CloudPro is always at the IMU centre, whatever you enter here.

Vehicle

- **Wheel constraints** follow the vehicle chosen above: the rear axle of a van or car may slide sideways at 0.10 m/s, of a truck at 0.15 m/s. For a platform without wheels (a boat, a trolley), switch **WHEELED VEHICLE CONSTRAINT** off on the Processing page, and **ZERO VELOCITY WHEN PARKED** too if it never stands still.
- **IMU TIME OFFSET** (ms): a fixed time shift of the IMU data, for expert use; leave it at 0 for the Certus.
- **REAR AXLE CENTRE**: the wheeled-vehicle constraint says the rear axle does not slide sideways or lift. If the Certus is not above the axle, enter where the axle centre is: in a turn at 0.3 rad/s each metre of offset is 0.3 m/s of sideways speed at the Certus. Measure it once per vehicle; a tape to 5 cm is enough. The mapper's rear-axle value is read from the capture when it is there.

IMU mounting correction

The wheel constraints need to know which way the vehicle points in the IMU's frame. When the alignment set in the Certus is a few degrees off how it is really mounted, PosPro estimates the difference (pitch and yaw) during every processing and writes it to the log (*vehicle frame against the IMU: pitch … yaw …*). The forward and backward passes can settle on different values, most often in yaw. Here you set them once:

- **PITCH** / **YAW OF THE VEHICLE AGAINST THE IMU** (degrees). Roll does not matter for the wheel constraints.
- **HOW PROCESSING USES THEM**: *Estimate them from the drive, starting at these values* (default; with values entered they may still move by about 1°), or *Use these values as they are* (held fixed).
- **USE THE VALUES FROM THE LAST PROCESSING** copies the estimate of the last processing (both passes weighted by their accuracy); the line under it shows that estimate and each pass's values.

> WHICH CAPTURES The correction belongs to captures recorded with the current Certus alignment. Once the alignment is corrected in the Certus, its output is already in the vehicle frame: new captures need 0°.

USE OFFSETS FROM THE CAPTURE (shown when the capture holds offsets) sets GPS1, GPS2, the Lidar offset, the rear axle and the vehicle from the mapper's `hardware.json`, or else from the Certus installation record (packet 185) and dual-antenna set-up (packet 196), and lists what it set. The line under it says where the offsets came from (*Offsets from: mapper hardware.json*, or *Offsets entered by hand*). The Installation step turns amber while the GPS1 offset is zero.

8 Step 4 - Processing

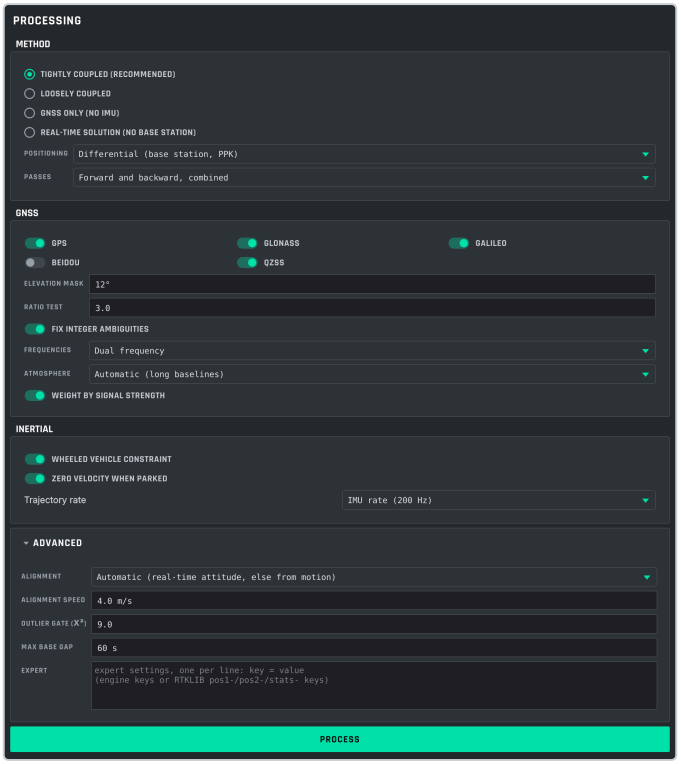


Figure 8.1 The Processing page with its default settings (tightly coupled, PPK, both passes) and the **ADVANCED** section opened.

Method

METHOD	WHAT IT DOES	NEEDS
TIGHTLY COUPLED (RECOMMENDED)	Raw carrier phase and code from rover (both antennas) and base go straight into the inertial filter. Keeps fixing with fewer than 4 satellites and repairs cycle slips with the IMU.	Raw IMU, raw GNSS, a base station with a position
LOOSELY COUPLED	GNSS positions are computed first (RTKLIB) and then blended with the IMU.	Raw IMU, raw GNSS; a base for PPK, or PPP without a base
GNSS ONLY (NO IMU)	The RTKLIB kinematic solution, forward and backward, combined. No attitude: roll and pitch are 0 and the heading follows the direction of travel. For comparison, or when there is no IMU data.	Raw GNSS; a base for PPK, or PPP
REAL-TIME SOLUTION (NO BASE STATION)	The Certus's own real-time trajectory (packet 20), cleaned in Trajectory Review. For when there is no base station, or RTK was used in the field.	The real-time solution in the captures

- **POSITIONING**: *Differential (base station, PPK)* or *Precise point positioning (no base)*. PPP works with loose coupling and GNSS only; tight coupling always needs a base. With PPP the trajectory is in ITRF2020 at the observation date, not NAD83(CSRS), and PosPro sets the project frame that way and warns.
- **PASSES**: *Forward and backward, combined* (the normal case: the two passes are combined by their accuracy, and their difference is the separation shown on the Quality page) or *Forward only (quick check)*.

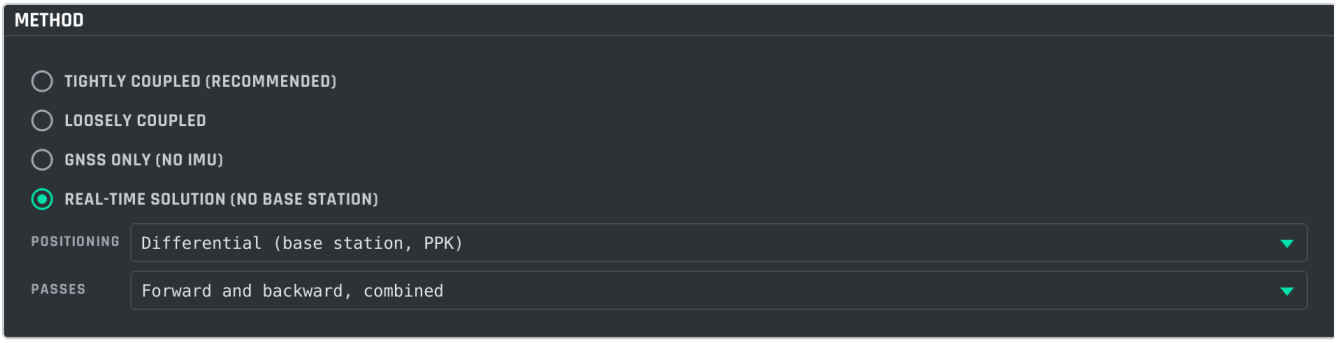


Figure 8.2 The real-time mode selected. It uses the Certus solution as recorded; the positioning and passes settings do not apply to it.

GNSS

- Constellations: `GPS`, `GLONASS`, `GALILEO`, `BEIDOU`, `QZSS` (GPS, GLONASS, Galileo and QZSS by default).
- `ELEVATION MASK` (12° by default): satellites lower than this are not used.
- `RATIO TEST` (3.0): how clearly the best integer ambiguity set must win before it is accepted as fixed. Higher is safer but fixes less.
- `FIX INTEGER AMBIGUITIES`: off gives a float solution only.
- `FREQUENCIES`: *Dual frequency* or *L1 only*.
- `ATMOSPHERE`: estimates the difference in ionosphere (per satellite) and troposphere between rover and base. *Automatic (long baselines)* turns it on beyond 10 km or 150 m height difference, where it keeps ambiguities fixable; on short baselines it only adds noise. *Always estimate*, *Off (short baselines)*.
- `WEIGHT BY SIGNAL STRENGTH`: weak signals (trees, buildings, multipath) count less than strong ones.

Inertial

- `WHEELED VEHICLE CONSTRAINT`: the vehicle does not slide sideways or jump; keeps the solution tight in GNSS gaps. How much sideways slip it allows comes from the vehicle on the Installation page (van and car 0.10 m/s, truck 0.15 m/s). Switch it off for a platform without wheels.
- `ZERO VELOCITY WHEN PARKED`: zero-velocity updates while the vehicle stands still.
- `TRAJECTORY RATE`: *IMU rate (200 Hz)*, 100, 50, 20 or 10 Hz.

Advanced

- `ALIGNMENT`: how the filter starts its attitude: *Automatic (real-time attitude, else from motion)*, *Real-time attitude from packet 20*, or *From motion (GNSS course, drive straight)*.
- `ALIGNMENT SPEED` (4.0 m/s): the speed needed to align from motion.
- `OUTLIER GATE (X²)` (9.0): GNSS measurements that disagree more than this with the filter are rejected.
- `MAX BASE GAP` (60 s): the longest interval between base observations that tight coupling bridges.
- `EXPERT`: extra settings, one per line as `key = value`. Keys starting with `pos`, `stats`, `misc` or `ant` go to RTKLIB; all others go to the inertial engine. Lines starting with `#` are ignored.

Process

Press `PROCESS`. PosPro first checks that the rover files are read and, for PPK and tight coupling, that a base is chosen (and asks when its position is only from the RINEX header). Then, in the background:

1. the rover files are read again if they changed;

2. for loose coupling, GNSS only and PPP: the GNSS solutions, forward and backward;
3. the inertial passes, forward and backward at the same time (for the real-time mode: the Certus solution as one pass);
4. the passes are combined; your Trajectory Review decisions (poor, interpolate) are applied again;
5. the trajectory is published for AlignPro and CloudPro.

The status line shows the stage and the progress bar the progress; `STOP` ends the run. The project box turns red and the PosPro workflow box reads *running*; another computer cannot process the same stage at the same time (*… is being processed by … on … since …*). When the run finishes, the Quality page opens. A run that fails or is stopped keeps the earlier results, and the PosPro box shows *failed*.

9 Step 5 - Quality

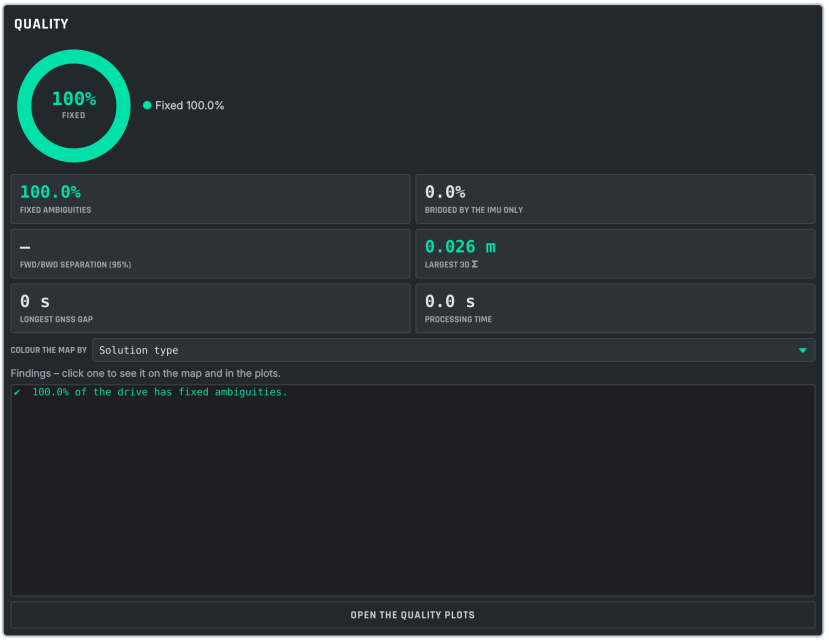


Figure 9.1 The Quality page for a real-time trajectory. With one pass there is no forward/backward separation; see the note below.

The donut shows the share of each solution type (Fixed mint, Float blue, PPP light blue, Code (DGNSS) amber, Single red, INS only violet). The cards:

CARD	GREEN	AMBER	RED
fixed ambiguities	≥ 95 %	≥ 75 %	below
bridged by the IMU only	share of the time without GNSS updates		
fwd/bwd separation (95%)	< 5 cm	< 20 cm	above
largest 3D σ	< 10 cm	< 50 cm	above
longest GNSS gap	the longest stretch bridged by the IMU only		
processing time	of the last run		

> SINGLE-PASS RESULTS With *Forward only* and in the real-time mode there is no backward pass to compare with, so the separation card shows –.

A good mobile-mapping trajectory is over 95 % fixed, has a forward/backward separation under 5 cm (95 %), and no 3D σ worse than 10 cm outside GNSS gaps. The Quality step is green from 95 % fixed, amber below.

Findings

A plain-language list: the share fixed, the longest GNSS gap bridged by the IMU and the number of gaps over 2 s, the forward/backward separation (95 % and maximum), up to five stretches where the passes disagree by more than 10 cm, up to three stretches where the estimated 3D accuracy is worse than 10 cm, and the IMU biases estimated at the end. Findings with a time show it in UTC; click one to mark that place on the map and zoom the quality plots to it (the plots open if they are closed).

The map

PosPro puts its layers in the group *PosPro: <project>* of the QGIS layer panel: **Real-time track** (what the Certus computed in the field, shown as soon as the files are read), **Base stations** and **Base distance rings**, **Trajectory** (one point per second, with the time, solution, satellites, height, attitude, speed, σ, separation and gap as attributes), **Trajectory line** and, after Trajectory Review, **Review windows**. `COLOUR THE MAP BY` colours the trajectory points by *Solution type*, *Estimated accuracy (3D σ)* or *Forward/backward separation* (classes < 2 cm, 2–5, 5–10, 10–30, 30–60, > 60 cm).

The quality plots

`OPEN THE QUALITY PLOTS` opens *PosPro – quality plots* in a window of its own (the same standard window as PosPro: floating by default, docked from its title-bar menu).



Figure 9.2 The quality plots, Ellipsoidal height tab: the two passes of the drive with the gap between them. The strip under the plot shows the solution type.

The tabs: **Position accuracy** (σ north, east, down), **Forward – backward separation** (north, east, down, where both passes have a solution), **Attitude accuracy** (σ roll, pitch, heading), **Satellites and solution** (satellites and the ambiguity ratio ÷ 10), **Speed**, **Ellipsoidal height**, **Roll and pitch** and **IMU bias estimates** (accelerometers in mg, gyros in deg/h).

Wheel to zoom, drag to pan, double-click (or `WHOLE DRIVE`) for the whole drive. Hover for the values at a time. A click marks the time with an orange line and shows that place on the map with an orange circle. With `LINK ZOOM` on, all tabs follow the same time range. Under every plot a strip shows the solution type in the map colours, and GNSS gaps (INS only) are shaded violet.

10 Step 6 - Trajectory Review

Trajectory Review finds where the trajectory misses your limits, lets you look at each place in the Trajectory editor, on the map and in the plots, and records what you decide for each. Decisions are kept with the project and carried to AlignPro and CloudPro.

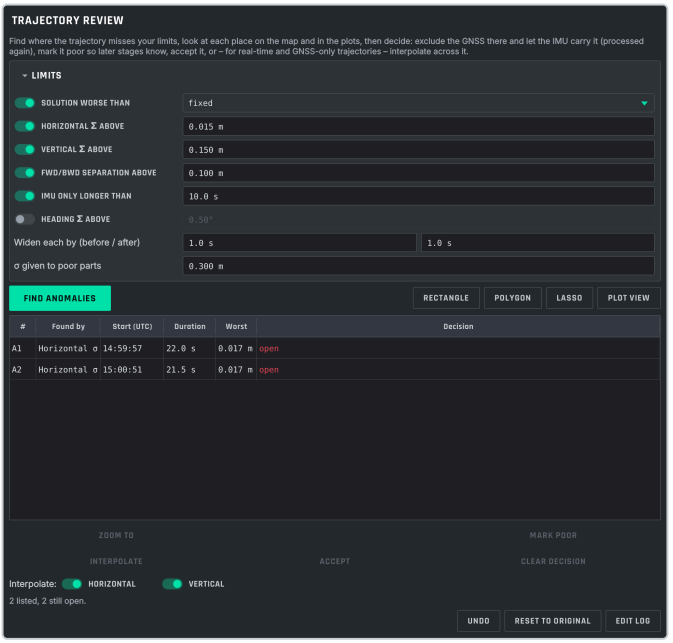


Figure 10.1 Anomalies found on a real-time trajectory with the horizontal σ limit set to 0.015 m. Both are still open.

Limits

LIMIT	DEFAULT	FINDS
SOLUTION WORSE THAN	fixed (on)	Samples worse than fixed, float, DGNSS or single (IMU-only samples are left to the IMU limit)
HORIZONTAL Σ ABOVE	0.100 m (on)	Horizontal accuracy worse than the limit
VERTICAL Σ ABOVE	0.150 m (on)	Vertical accuracy worse than the limit
FWD/BWD SEPARATION ABOVE	0.100 m (on)	The passes disagreeing horizontally by more than the limit
IMU ONLY LONGER THAN	10.0 s (on)	GNSS gaps longer than the limit
HEADING Σ ABOVE	0.50° (off)	Heading accuracy worse than the limit
WIDEN EACH BY (BEFORE / AFTER)	1.0 s / 1.0 s	A margin added to each window, because a drop in quality is often reported a little late
Σ GIVEN TO POOR PARTS	0.300 m	The σ that MARK POOR gives

Flagged samples less than 1.5 s apart join into one window.

Finding and selecting

- `FIND ANOMALIES` checks the trajectory as processed (before any review edit) against the limits and lists the windows; a message says how many were found. Run it again after changing the limits.
- `RECTANGLE` , `POLYGON` , `LASSO` : select part of the trajectory on the QGIS map. Rectangle: drag. Polygon: click the corners; right-click or double-click to finish. Lasso: drag around the part. Esc cancels. The parts of the trajectory inside the shape are added to the list as *Selected by hand*: parts less than 30 s apart become one window, and a drawing that catches more than 20 separate parts becomes one window from the first to the last. The same window drawn twice is listed once.
- `PLOT VIEW` : adds the time range the quality plots are zoomed to. Open the plots and zoom first.

Processing finds the anomalies by itself with the limits above, so the list is ready when processing ends. Hand selections belong to the trajectory they were drawn on: processing again takes them off the list (decisions already made on them stay). `REMOVE FROM LIST` takes the selected **S** rows off, and `CLEAR HAND SELECTIONS` takes them all off.

`SHOW` narrows the list to the open windows, the ones found by the limits (A), the hand selections (S) or the decided ones; click a column heading to sort by it (again to reverse) – for example by *Worst* to look at the largest first. To get rid of old hand selections: `SHOW` › *Selected by hand*, select them (Ctrl+A) and `REMOVE FROM LIST` , or `CLEAR HAND SELECTIONS` .

The list shows the window (**A** found, **S** selected by hand), what found it, the start (UTC), the duration, the worst value and the decision (*open* in red until you decide). The windows are drawn on the map in the layer *Review windows*: red open, blue GNSS excluded, amber poor, violet interpolated, mint accepted. Double-click a row, or `ZOOM TO` , to see it on the map and in the plots.

The Trajectory editor

`OPEN TRAJECTORY EDITOR` (next to `FIND ANOMALIES`) opens the processed trajectory in its own window: the **plan view** on top, the **height profile** (height against time) below, and on the right the **sections** of the job and the **review list**. Everything you decide here is the same decision as on the Review page, so both always agree.

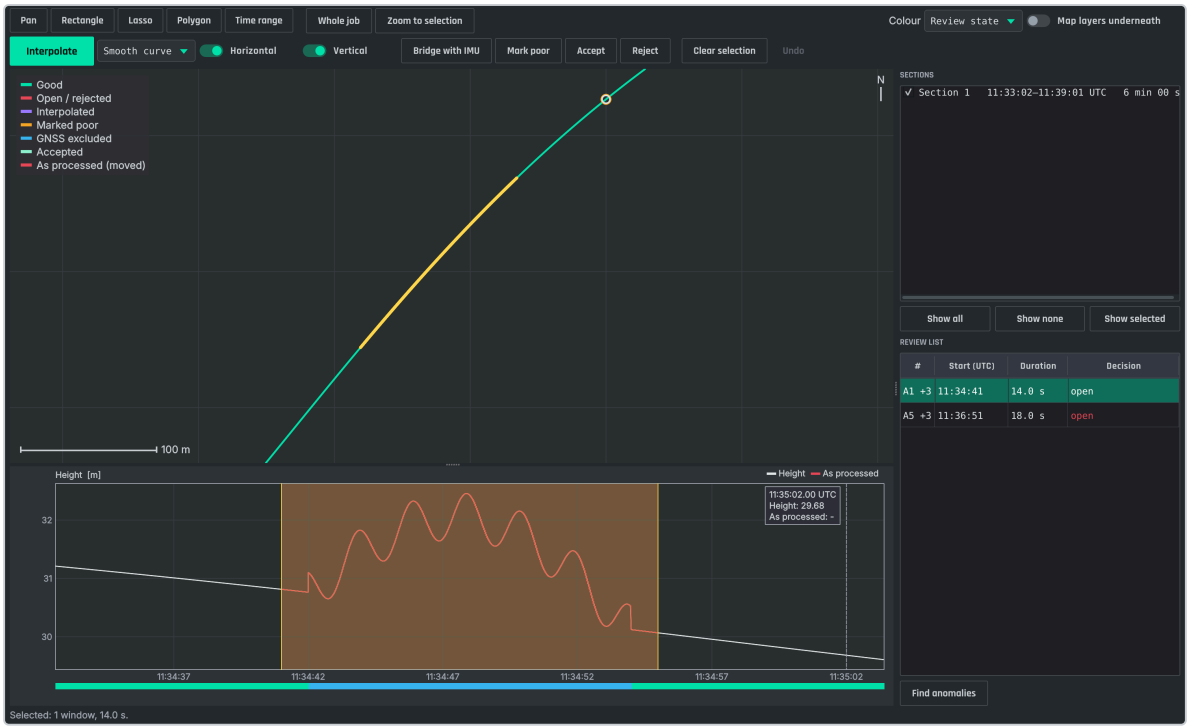


Figure 10.2 The Trajectory editor on an anomaly: the window chosen in the review list is selected (yellow) in the plan and in the profile, where the height as processed (red) jumps about. The band under the profile shows the solution type (mint fixed, blue float).

The views are linked. Move the mouse over the plan and the profile shows a line at that moment with its height; move it over the profile and a ring marks the same place in the plan. Click a point in either and it is marked in both. Zoom or pan the plan (wheel, drag) and the profile shows the time of what is in view; zoom the profile and the plan follows when that stretch is off screen. The line under the views gives the time, position, height, solution type and review state under the mouse.

TOOL	USE
PAN	Drag to move, wheel to zoom, click a point to mark it in both views, double-click to see the whole job. The middle mouse button pans with any tool.
RECTANGLE, LASSO, POLYGON	Select bad points in the plan view (polygon: click the corners, right-click or double-click to finish; Esc cancels). The selection is shown in both views; parts less than 3 s apart become one window.
TIME RANGE	Drag across the profile to select a time range, e.g. a height spike.
WHOLE JOB, ZOOM TO SELECTION	Fit both views to the shown sections or to the selection.
COLOUR	*Review state*: mint good, red open or rejected, violet interpolated, amber poor, blue GNSS excluded, pale mint accepted. *Solution type*: the mapper's solution colours. Positions the edits moved are also drawn as they were processed, in faint red.
DIRECTION	Chevrons along the track pointing the way the vehicle drove (on by default).
ANTENNAS, CHECK ANTENNAS	ANTENNAS draws GPS1 (mint) and GPS2 (blue) along the track from the Installation offsets, and at the point under the mouse or clicked, the vehicle: the IMU joined to both antennas. After CHECK ANTENNAS (see *Installation › Checking the antennas*) the GPS2 positions it measured are drawn as orange dots: on the blue line when all is well, on the other side of GPS1 when the antennas are swapped.
MAP LAYERS UNDERNEATH	Draws the layers of the QGIS map (your basemap, roads) under the plan view.

Sections	The job split where the vehicle stood still for 30 s or more, or the data stops. Untick sections to hide them; **SHOW SELECTED** shows only the highlighted ones; double-click a section to zoom to it.
Review list	One row per place (a place found by several limits shows as e.g. *A1 +3*). Click a row to select its window in both views and zoom there.

With a selection, choose what to do with it (the buttons above the views):

BUTTON	EFFECT
INTERPOLATE	Replaces the selected positions between the good points on each side, **HORIZONTAL** and/or **VERTICAL**. *Smooth curve* leaves and meets the good points at their own speed and direction, so it follows a bend; *Straight line* joins them directly (good for a height spike on level ground). A selection at the very start or end of the trajectory is carried on from its one good side at that point's speed and direction.
BRIDGE WITH IMU	The same as **EXCLUDE GNSS**: the IMU carries the trajectory there after **PROCESS AGAIN** (shown when waiting). Tightly and loosely coupled only.
MARK POOR, **ACCEPT**	As on the Review page.
REJECT	Marks it bad, to fix later: it stays *open* in red.
UNDO	Takes the last decision back.

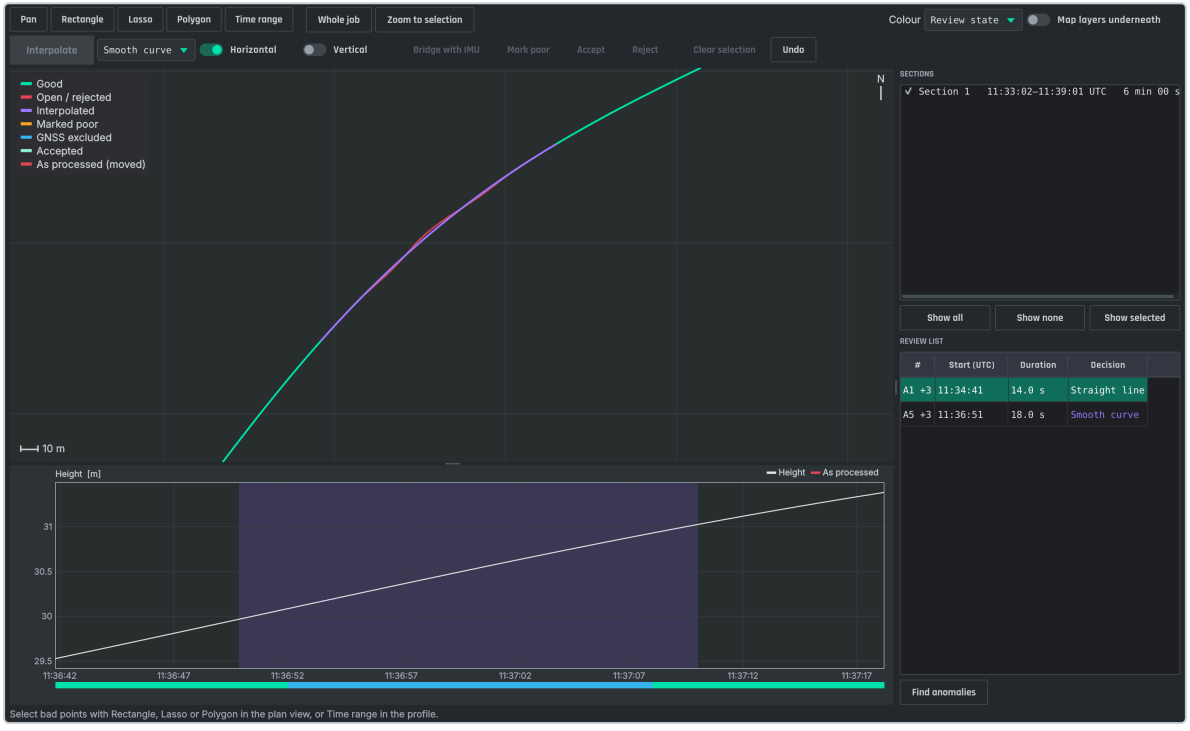


Figure 10.3 A sideways jump fixed with a smooth curve, horizontally only: the trajectory now (violet) follows the bend, and the positions as processed are drawn in faint red beside it.

> INTERPOLATE OR BRIDGE WITH THE IMU? Interpolation is best for short stretches: spikes, a few seconds of wrong fixes. Over a longer stretch the IMU knows the turns and bumps that a curve cannot guess, so **BRIDGE WITH IMU** usually follows the road better; PosPro reminds you when you interpolate more than 10 s of a coupled trajectory.

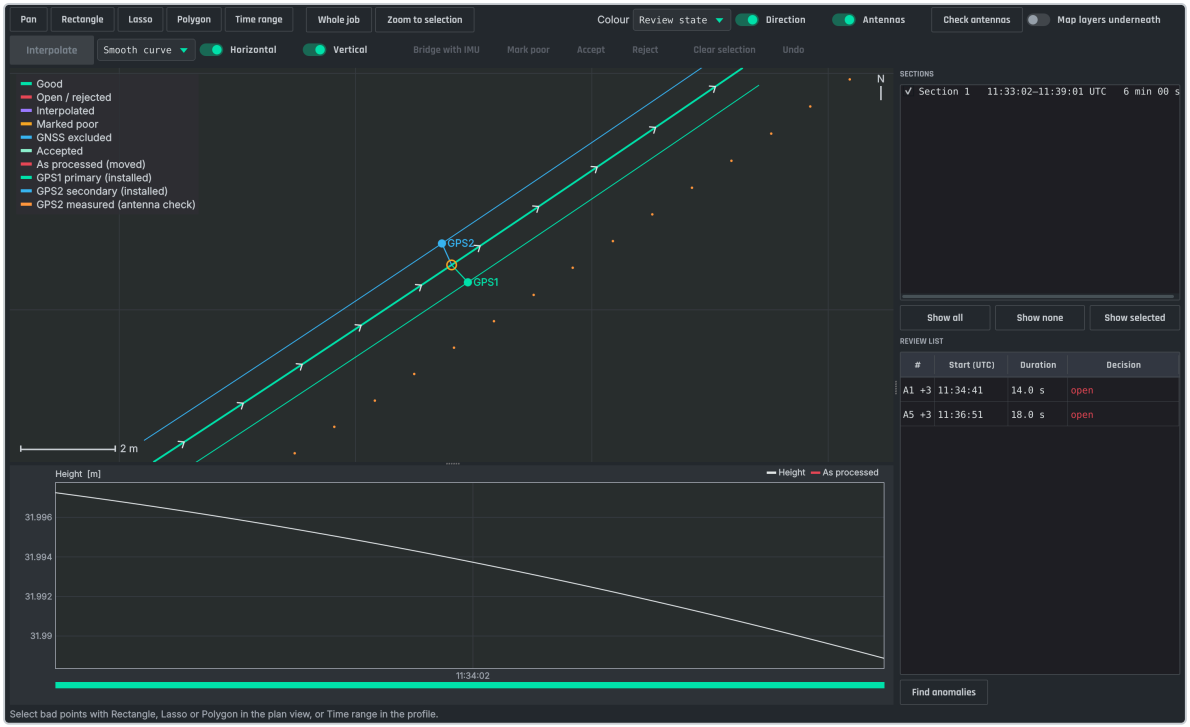


Figure 10.4 Direction chevrons, the antennas along the track and the vehicle at the clicked point, with the GPS2 positions measured by the antenna check (orange): they lie on the starboard side of GPS1, not on the blue GPS2 line, so the antennas were swapped.

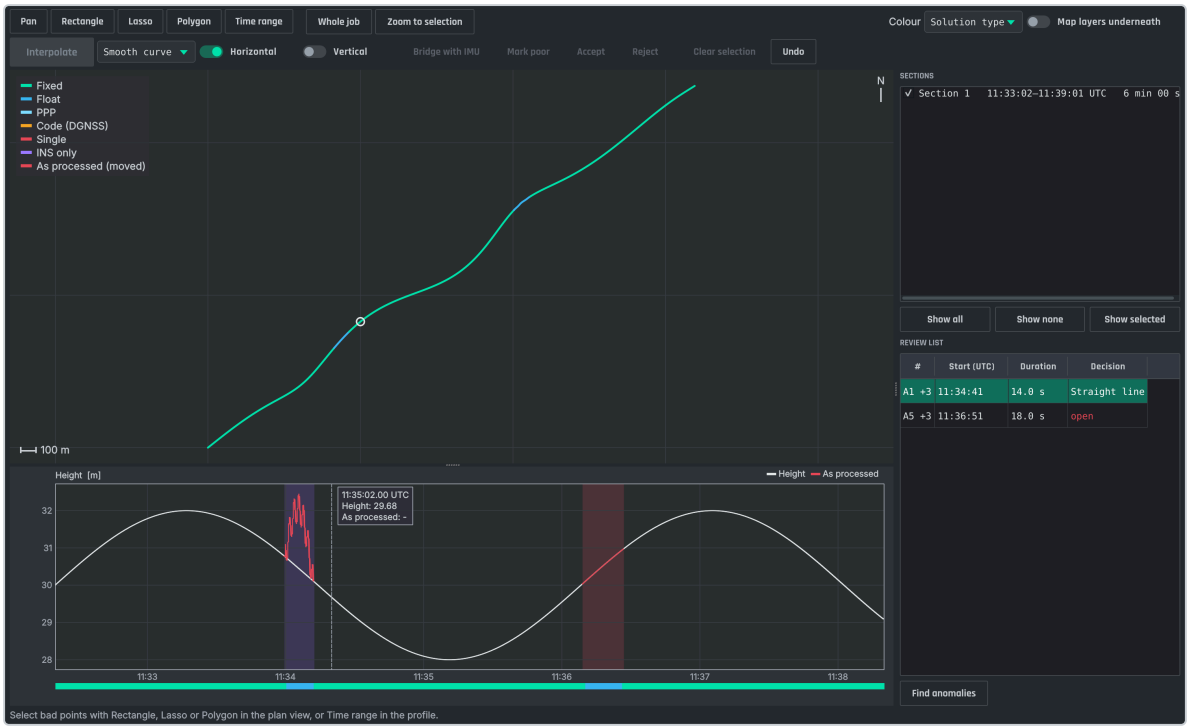


Figure 10.5 The whole job coloured by solution type: the height spike is interpolated (violet band), the second place is still open (red band).

Deciding

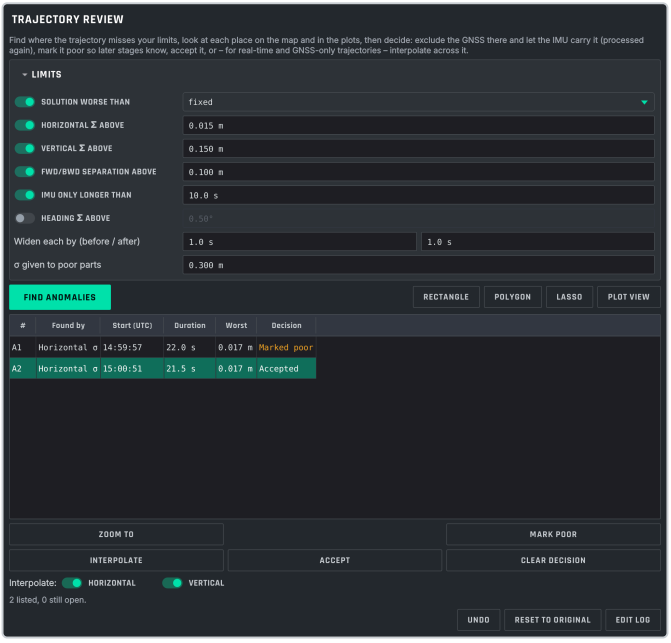


Figure 10.6 The same windows after the decisions: A1 marked poor, A2 accepted.

Select one row or several, then:

DECISION	EFFECT	APPLIES
`EXCLUDE GNSS`	GNSS is not used in these windows; the IMU carries the trajectory, as through a tunnel. Use it where multipath or a wrong fix pulled the trajectory away. Tightly and loosely coupled only.	After `PROCESS AGAIN`; a note in amber says the exclusions are waiting, and the Review step reads *process again*
`MARK POOR`	The trajectory is kept, but its position σ is raised to at least `Σ GIVEN TO POOR PARTS`, in the SMRMSG file, on the Quality page and in the published trajectory, so later stages know not to trust it.	At once
`INTERPOLATE`	Replaces the positions between the good samples on both sides, as a *Smooth curve* or a *Straight line*, `HORIZONTAL` and/or `VERTICAL`, in any processing mode (see *The Trajectory editor*). For longer stretches of a coupled trajectory, `EXCLUDE GNSS` usually follows the road better.	At once
`ACCEPT`	Looked at and fine as it is.	At once
`CLEAR DECISION`	The window is open again.	At once

Decisions never change the processed result itself: PosPro keeps an untouched copy and applies your decisions to it, so `UNDO` (the last change), `RESET TO ORIGINAL` (every decision removed, after a question) and processing again lose nothing. After a decision that changes the trajectory, the Quality page, the map and the plots are updated and the trajectory is published again. `EDIT LOG` lists every detection and decision with its time.

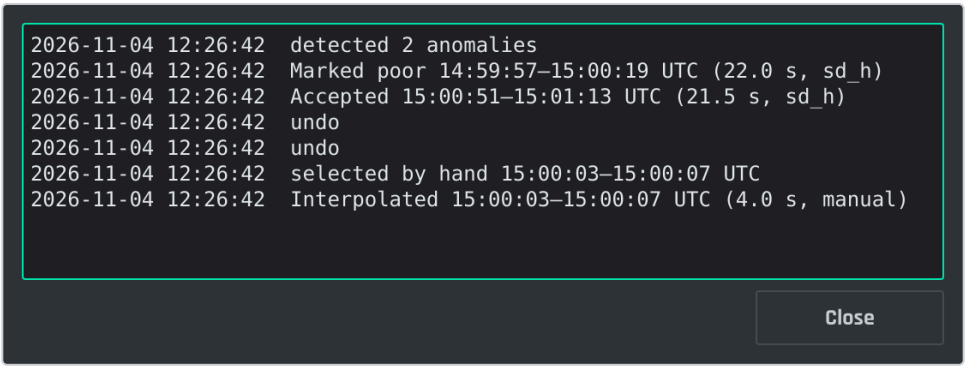


Figure 10.7 The Trajectory Review log.

> WINDOWS INSIDE WINDOWS Deciding on a window that lies inside a larger window that already has a decision adds a decision for the smaller window; the larger window keeps its own. To change a decision, select that window itself and decide again, or use `CLEAR DECISION`.

The Review step shows *find anomalies* before the first search, *N open* in amber while windows are undecided, *process again* while GNSS exclusions wait, and *N reviewed* in green when all are decided.

> NO BASE STATION? Choose `REAL-TIME SOLUTION (NO BASE STATION)` on the Processing page. PosPro then uses the trajectory the Certus computed in the field (RTK when it had corrections), and Trajectory Review cleans it the same way, with the Trajectory editor for the parts it got wrong.

11 Step 7 - Calibration

The lever-arm solver estimates the IMU-to-antenna offsets from the drive itself and shows them beside the ones you entered, so a mis-measured antenna shows up before it spoils a survey. It works on data you have already processed; measure the offsets as well as you can first, and let the solver check and refine them.

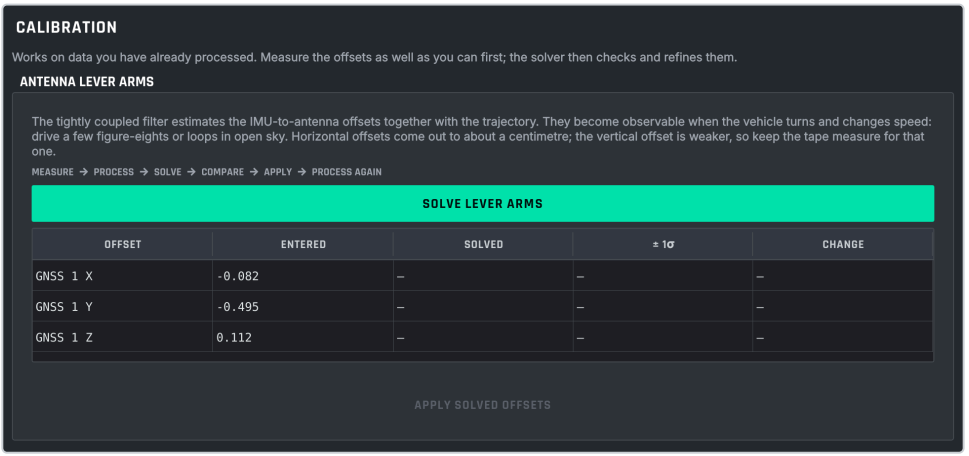


Figure 11.1 The Calibration page before solving: the entered GPS1 offsets, nothing solved yet.

1. Process the drive with **TIGHTLY COUPLED** and a base station, fixed for most of the drive. The solver needs tight coupling; in any other mode it says so (Figure 11.2).
2. Press **SOLVE LEVER ARMS**. Both passes run again with the offsets as unknowns, about as long as a normal processing run, and are combined by their accuracy.
3. Read the table: **OFFSET** (GNSS 1 and, when used, GNSS 2; X forward, Y right, Z down), **ENTERED**, **SOLVED**, **± 1Σ** and **CHANGE**. A change in amber is larger than three standard deviations and 2 cm.
4. Re-measure an amber offset if you can. Otherwise press **APPLY SOLVED OFFSETS**: the solved values go to the Installation page. Process again so the trajectory uses them, and enter the same values in the Certus and the mapper Settings if you keep them.

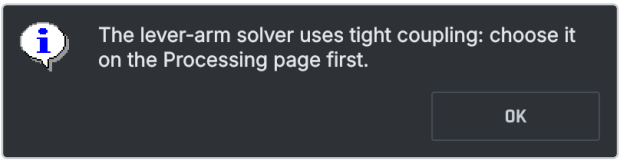


Figure 11.2 The solver needs tight coupling.

The offsets become observable when the vehicle turns and changes speed: drive a few figure-eights or loops in open sky. Horizontal offsets come out to about a centimetre; the vertical offset is weaker, so keep the tape measure for that one. A straight drive cannot separate the forward offset from the IMU's own errors.

12 Step 8 - Export

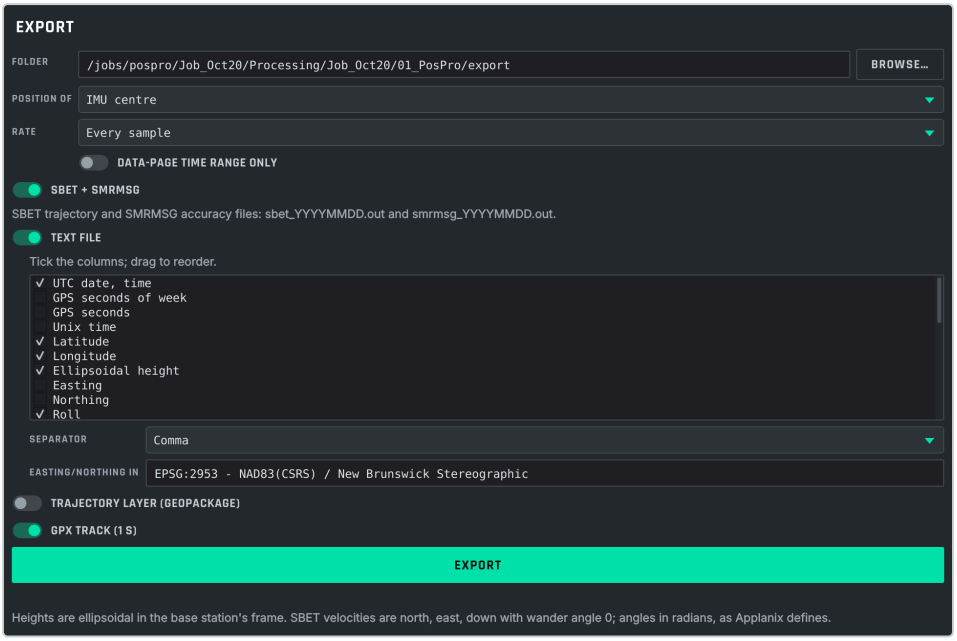


Figure 12.1 The Export page with SBET + SMRMSG, a text file with seven columns and a GPX track chosen.

SETTING	WHAT IT DOES
FOLDER	Where the files go; by default `01_PosPro/export` in the project. **BROWSE…** opens a folder picker.
POSITION OF	*IMU centre* (the Certus reference point), the output point from the Installation page (by its name, when its offset is not zero), or *GNSS antenna* (GPS1).
RATE	Every sample, 200, 100, 50, 10 or 1 Hz.
DATA-PAGE TIME RANGE ONLY	Export only the time range set on the Data page.
SBET + SMRMSG	The trajectory and its accuracy: `sbet_YYYYMMDD.out` and `smrmsg_YYYYMMDD.out`.
TEXT FILE	`<project>_YYYYMMDD.txt` with the columns you tick, in the order you drag them to: UTC date, time; GPS seconds of week; GPS seconds; Unix time; Latitude; Longitude; Ellipsoidal height; Easting; Northing; Roll; Pitch; Heading; Velocity N, E, D; Speed; σ North, East, Down; σ Roll, Pitch, Heading; Solution type; Satellites; Separation H, V. **SEPARATOR**: Comma, Tab, Space or Semicolon. **EASTING/NORTHING IN**: a transverse Mercator, UTM or stereographic system (EPSG:2953, NAD83(CSRS) / New Brunswick Stereographic, by default).
TRAJECTORY LAYER (GEOPACKAGE)	`<project>_trajectory.gpkg`: the Trajectory map layer (one point per second with its quality attributes).
GPX TRACK (1 S)	`<project>_trajectory.gpx`: the trajectory at the Certus reference point, one point per second, with ellipsoidal heights.

Press **EXPORT**; the files written are listed under the button, and the Export step turns green (*exported* and the time) until the trajectory is processed again. Every export also writes `<project>_export_note.txt`, which records the project, when and how it was processed, the frame and epoch, the base and its position source, the heights, the time system, the exported point and its offset, and the attitude frame. Keep it with the files.

File formats

FILE	CONTENT
`sbet_YYYYMMDD.out`	17 little-endian doubles per record: GPS seconds of week, latitude, longitude (rad), ellipsoidal height (m), velocity north, east, down (m/s), roll, pitch, heading (rad), wander angle (0), body accelerations, body angular rates.
`smrmsg_YYYYMMDD.out`	10 doubles per record: GPS seconds of week, σ north, east, down (m), σ velocity north, east, down (m/s), σ roll, pitch, heading (arc-minutes).
`<project>_YYYYMMDD.txt`	One header line with the column names, then one line per sample.

Heights are ellipsoidal, in the frame of the base station's position (for PPP: ITRF2020 at the observation date; for the real-time mode: the frame of the Certus real-time solution, that is of the RTK corrections used in the field). Apply the geoid in the point-cloud processing. SBET velocities are north, east, down with wander angle 0; angles are in radians. Use the same reference point in the point-cloud processing as in the export.

What PosPro hands to AlignPro and CloudPro

At the end of every processing run, and after every review decision that changes the trajectory, PosPro publishes the trajectory in `01_PosPro/out/`. You do not need the Export page for AlignPro and CloudPro: they read these files.

FILE	CONTENT
`trajectory_sbet.out`	SBET at the Certus reference point (IMU centre), every sample
`trajectory_smrmsg.out`	SMRMSG for the same samples (σ raised where you marked parts poor)
`trajectory_qc.csv`	The 1 Hz quality table (position, attitude, velocity, σ, separation, solution type, satellites, gap, IMU biases)
`trajectory.json`	The description of the trajectory, written last (schema `dcs.trajectory/1`)

TRAJECTORY.JSON

FIELD	HOLDS
`schema`, `id`, `producer`, `created`, `project`	`dcs.trajectory/1`; a content id of the data files (the later stages use it to see whether they are out of date); PosPro and engine versions; time; project name
`built_from`	The captures (by fingerprint) and the shared settings the trajectory was made from
`files`	`sbet`, `smrmsg`, `qc`
`time`	System GPS, GPS week, leap seconds, start and end (GPS seconds), SBET time as GPS seconds of week
`frame`	Frame name and epoch (that of the base position); for a real-time trajectory, the frame of the Certus's RTK corrections; heights ellipsoidal
`reference_point`, `body_frame`	Certus reference point (IMU centre), offset 0, 0, 0; Certus aligned vehicle frame: X forward, Y starboard, Z down; roll + starboard down, pitch + bow up, heading + clockwise
`levers_m`	GNSS 1 and GNSS 2 offsets (with their source, and whether GNSS 2 was used), the Lidar offset from the capture's `hardware.json`
`rate_hz`	The rate of the published SBET, measured from its time stamps
`quality`	Method, positioning, % fixed, % IMU only, separation (95 %), largest horizontal and vertical σ
`base_station`	The base used

`review`	The Trajectory Review decisions (GNSS excluded, reduced quality with its σ, interpolated, accepted windows) and the limits used
`attitude`	For GNSS-only trajectories only: attitude not observed (roll and pitch 0, heading from the direction of travel)

When you change something the trajectory depends on (the coordinate frame, the base station, the antenna offsets) or a capture changes, the PosPro box turns ⚠ **stale** in every plugin, and AlignPro and CloudPro show that their results are older than the trajectory. Process again to bring the chain up to date.

13 Troubleshooting

Messages

MESSAGE	WHAT TO DO
add the rover files (.anpp or .clraw) first / rover file not found: …	Add the files on the Data page; if the job folder moved, use **RAW DATA MOVED? RELOCATE THE JOB FOLDER…** .
no raw IMU (packet 28) with valid time was found / no raw satellite observations (packet 60) were found	The Certus did not record these packets. Only the real-time mode can run on these captures. Set the Certus outputs on the mapper for the next capture.
Read the rover files (Data). / Add a base station (Base stations).	Shown by **PROCESS** when a step before it is missing.
the base station … has no position yet	Enter or compute the base position on the Base page.
tight coupling needs a base station; for PPP choose "Loosely coupled"	PPP works with loose coupling or GNSS only.
no navigation (ephemeris) data: connect to the internet or add a RINEX navigation file on the Data page	Turn on **DOWNLOAD BROADCAST EPHEMERIS** with internet, or add a navigation file.
navigation file …: N orbits disagree with the INS's own and the IGS broadcast orbits … and are left out (…)	That file holds wrong orbits for those satellites. PosPro already left them out; take the file out of the base folder so it is not used again.
the solution ran away: its σ reached … km	Processing stopped: the trajectory cannot be used, and Trajectory Review cannot repair it. Check the warnings above it: a bad navigation file, a wrong base position (compare the PPP result with the RINEX header position or CSRS-PPP), or IMU data that did not arrive in full. Fix the cause and process again.
the largest σ is … m	A warning: look at the quality plots before using the trajectory.
broadcast ephemeris for … not downloaded (…)	A warning: PosPro goes on with the files you supplied and the rover's GPS ephemeris.
no precise orbit/clock found for …	Final products appear about two weeks after the day, rapid ones after about a day. Try again later, or use CSRS-PPP.
antenna calibrations (igs20.atx) not available	Base antenna phase-centre variations are not applied; connect to the internet once so the file is cached.
the base files … list different observation types; add them as separate base stations	Add each file as its own base.
the base file is only … h long	PPP is good to a few cm only after 2 h; use a longer file or CSRS-PPP.

CSRS-PPP needs the e-mail address you use as your NRCan user name	Enter it under **E-MAIL**.
the rover files hold no real-time solution (packet 20)	The real-time mode cannot run on these captures.
GPS2 could not be measured (no GPS2 observations, baseline did not fix)	Record a few minutes with both antennas in the open, or enter GPS2 by hand.
the lever-arm solver needs tight coupling / the filter did not report lever arms	Process tightly coupled with a base, with most of the drive fixed.
Easting/northing: choose a transverse Mercator, UTM or stereographic system	Choose such a system on the Export page.
process the project first	Export needs a processed trajectory.
… is being processed by … on … since …	Another QGIS is processing this stage of the project. Wait for it to finish.
…% of the IMU samples are missing (… gaps, the longest … s): the INS stream did not arrive in full	The recording has holes. PosPro bridges short gaps; for the next capture record with mapper 2.10.1 or later and lower the output rates (Recording settings).
the IMU's aligned frame is … deg (pitch) / … deg (yaw) off the vehicle	The INS alignment set in the Certus does not match how it is mounted. PosPro allows for it (the offset is estimated while driving); correct the alignment in the Certus so its real-time solution benefits too.
The processing engine is missing	Reinstall PosPro. On Windows, allow `pospro-engine.exe` if your security software asks.
The user guide is missing from this installation; reinstall PosPro.	Reinstall PosPro.

Results

WHAT YOU SEE	LIKELY CAUSE	WHAT TO TRY
Fixed below 75 %	Base far away (over 40 km), poor sky	A nearer base; **ATMOSPHERE** on Automatic or Always; a higher elevation mask
Large separation at the start or end only	Vehicle parked: heading cannot be resolved without motion	Drive a few turns before and after the survey; use both antennas
Large separation in a long gap	IMU drift both ways	Expected; check the σ plot over the gap; the wheeled-vehicle constraint and the rear axle offset help
Everything offset by the same amount	Base position from the RINEX header, or the wrong frame or epoch	Set the base position (CSRS-PPP)
Trajectory tilts or swings in turns	Wrong antenna offset or rear axle offset	Check the Installation drawing; solve the lever arms

14 Reference

Defaults of a new project

SETTING	DEFAULT
Frame	NAD83(CSRS), epoch 1997.0 (until a base position sets it)
Method / positioning / passes	Tightly coupled / PPK / forward and backward
Constellations, mask, ratio	GPS, GLONASS, Galileo, QZSS; 12°; 3.0
Ambiguities, frequencies, atmosphere, SNR weighting	Fixed; dual; automatic; on
Wheeled vehicle constraint, zero velocity, rate	On; on; IMU rate
Alignment, speed, gate, max base gap	Automatic; 4 m/s; χ^2 9; 60 s
IMU, vehicle	Certus Evo; Van (or the vehicle recorded in the capture)
Export	IMU centre, every sample, SBET + SMRMSG
Review limits	See Trajectory Review

Folders

PATH (IN `PROCESSING/<NAME>/01_POSPRO/`)	HOLDS
`pospro.json`	PosPro's settings, bases, results summary and review decisions
`work/extract/`	The data read from the captures (IMU, observations, real-time track, coverage, hardware record)
`work/base/`	Downloaded and prepared base files, base PPP
`work/gnss/`, `work/ins/`	GNSS solutions, inertial passes and their settings
`work/results/`	The combined trajectory, its quality table and their untouched copies for Trajectory Review
`out/`	The published trajectory for AlignPro and CloudPro
`export/`	Exports (default folder)

Recording settings

The first field test (3 Nov 2026, mapper 2.9.0) lost 31 % of the Certus stream in 357 gaps of up to 1.4 s. PosPro bridges short gaps, and the Data step reports how much IMU data is missing, but a clean recording is always better. Two causes add up: the mapper's computer, busy recording the Lidar, could drop network packets, and the Certus stream then stalls and the Certus throws data away while it waits (mapper 2.10.1 reduces that load); and the outputs asked for about 69 KB/s, close to what came through. When the Data step reports missing IMU samples, record with mapper 2.10.1 or later and lower the output rates:

PACKET	WHAT POSPRO USES IT FOR	SETTING
20 System state	time for every IMU sample, real-time solution	5 ms (200 Hz), keep

28 Raw sensors	the IMU data	5 ms (200 Hz), keep
60 Raw satellite data	GNSS observations of both antennas	1000 ms (matches a 1 s base); 500 ms if the check below stays clean
61 Ephemeris; 185, 196 offsets	orbits; installation and dual-antenna offsets	as the Certus sends them
29 Raw GNSS fix	not used for post-processing	100 ms or off
26 Euler standard deviation	not used	off

With these settings the stream is about 37 KB/s. Check after the next drive: the Data step should report under 1 % missing IMU samples.

Field practice that improves the trajectory

- Record raw IMU and raw GNSS of both antennas on the Certus with GLONASS on (it helps the fix rate), within what the port can carry (Recording settings).
- Log a base at 1 s, within 20 km, for the whole drive plus 10 minutes either side. Park 1–2 minutes, then drive a few turns at the start and the end; this resolves heading and IMU biases for both passes.
- Measure the antenna offsets to the millimetre in the Certus aligned frame, and the rear axle to 5 cm; then check them with the lever-arm solver.